



Report on

RELIABILITY PREDICTION

OF

Camera Model No.: HC-IPC-TI4312DVZK-N8-O
(CCTV CAMERA)

as per Telcordia SR-332, Issue 4



CENTRE FOR RELIABILITY,
STQC Directorate
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Ministry of Electronics and Information Technology (MeitY)
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1. This report refers only to the sample actually subjected to Design Reliability Analysis.
2. The analysis report provided is solely based on a theoretical estimation for predicting design reliability.
3. The accuracy and validity of the design reliability prediction cannot be guaranteed due to its theoretical nature.
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1.0 SCOPE

- 1.1 Service request no. : 2602311
- 1.2 Name & Address of the Client : Hi-Focus Electronics India Pvt Ltd
3, Sunguvar Agraharam St,
Chindatripet, Chintadripet, Chennai,
Tamil Nadu 600002.
- 1.3 Description : **CCTV CAMERA**
Make : Hi-Focus Electronics India Pvt Ltd
Model No. : HC-IPC-TI4312DVZK-N8-O
Serial No. : HFP25100543120730
- 1.4 Project : Received Date : 11-05-2026
Completed Date : 09-06-2026
- 1.5 (a) Standard/ Analysis / Test Procedure Ref. : Telcordia SR-332, Issue 4
(b) Scope : Reliability Prediction by Parts Count Method



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2.0 INTRODUCTION

This report presents the results of Reliability Prediction Analysis performed on **Camera Model No.: HC-IPC-TI4312DVZK-N8-O** as per Telcordia SR-332, Issue 4 as depicted in Figure 1.0, 2.0, 3.0, 4.0, 5.0, 6.0. and 7.0 This exercise was performed under **Ground, Fixed, Controlled (GB)** environment as requested by **Hi-Focus Electronics India Pvt Ltd.**

Design Reliability Prediction is the process of quantitatively assessing whether a proposed or actual equipment design will meet a specified reliability requirement. Design Reliability Prediction has various applications in the design and development process including substantiating quantitative design comparisons with competitive products, selecting warranty periods, advertising and marketing. It provides guidance, relative to expected inherent reliability of a given design.

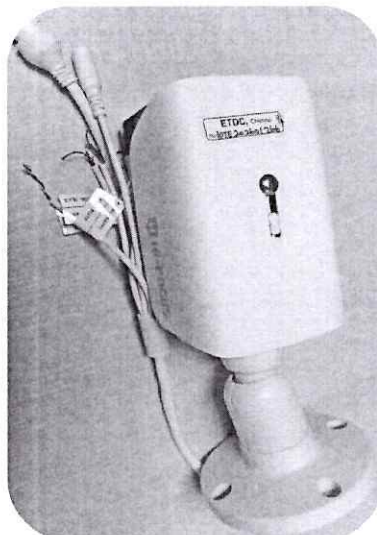
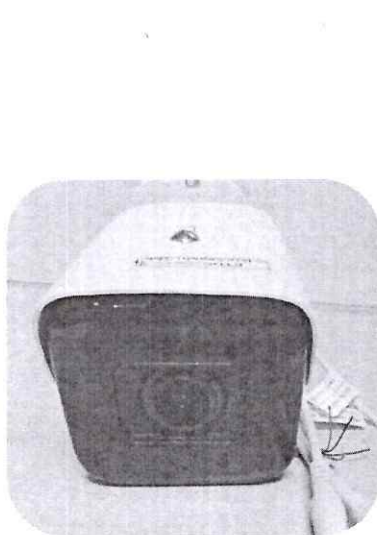


Figure 1.0 (a & b) Front & Top view of camera



Figure 2.0 (a) Markings on camera

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Camera Model No.: HC-IPC-TI4312DVZK-N8-O

Date of Issue
09-06-2026

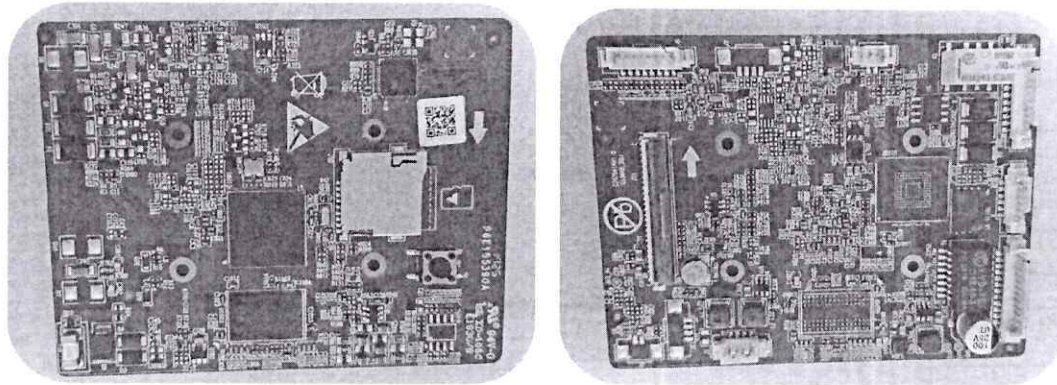


Figure 3.0 (a & b) Front & Rear View of PCB 1

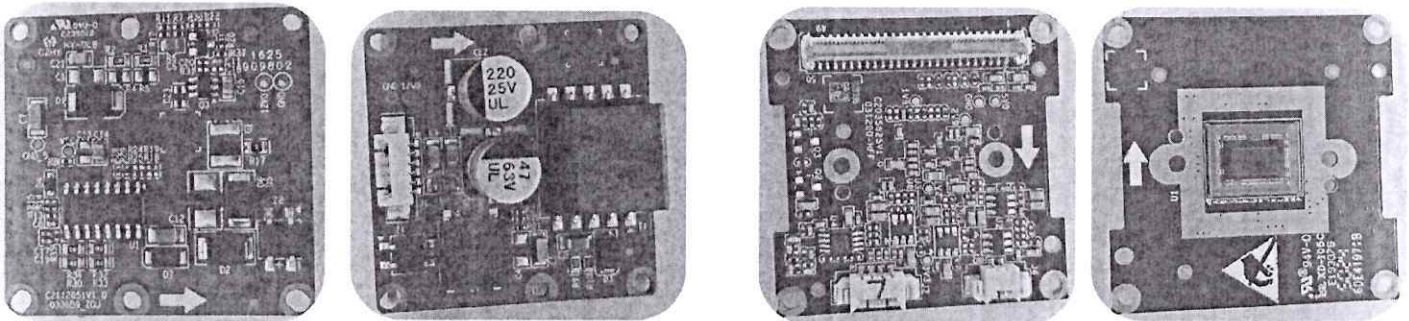


Figure 4.0 (a & b) Front & Rear View of PCB 2

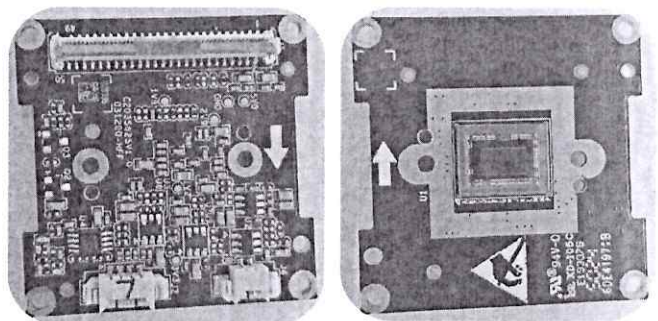


Figure 5.0 (a & b) Front & Rear View of PCB 2

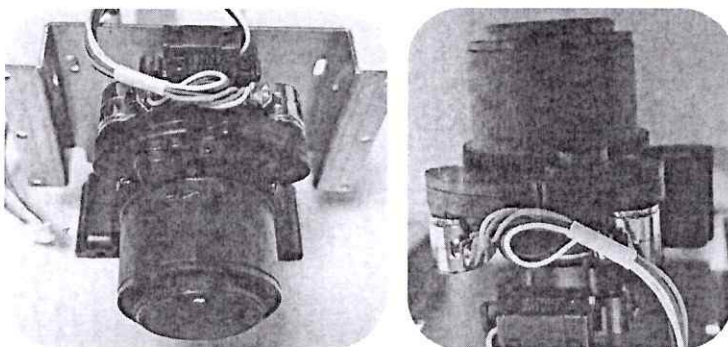


Figure 6.0 (a & b) Motorized Lens

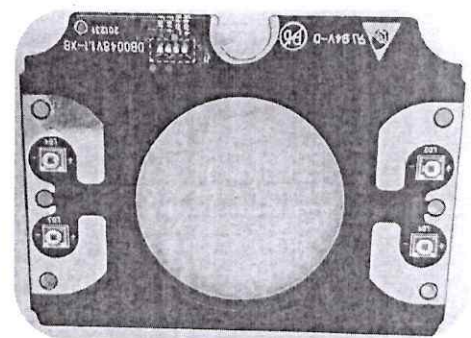


Figure 7.0 (a) IR Board & LEDs

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3.0 APPLICABLE DOCUMENTS AND TOOLS

Telcordia SR-332, Issue 4 - Reliability Prediction Procedure for Electronic Equipment

4.0 RELIABILITY PREDICTION OBJECTIVES

Objectives of this analysis are to estimate the Mean Time between Failure (MTBF) of **Camera Model No.: HC-IPC-TI4312DVZK-N8-O** using Telcordia SR-332, Issue 4 and to obtain quantitative design information as applicable. The main objectives are:

- a. Provide an estimate of the steady-state failure rates for device design based on generic reliability parameters and obtain feasibility evaluation.
- b. Provide feedback for designers.
- c. Identify potential problem areas during design review.
- d. Provide MTBF of the system and sub-assemblies and failure rate of system/sub-assemblies/ Components.

5.0 SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method - ASSUMPTIONS

Reliability Prediction using SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method for reliability prediction operates under the following assumptions: assumes a serial relationship between parts of the system, in which, failure of any part constitutes system failure. The model also assumes that

- a. At the device level, it is assumed that no detailed reliability data is available. Predictions are based on generic reliability parameters.
- b. Predictions are made using generic reliability parameters as specified in the relevant sections of SR-332, which are applied to estimate steady state failure rates.
- c. At the unit level, failure rate predictions are based solely on the failure rates of the individual devices that compose the unit, without considering interactions between devices.
- d. This method primarily estimates steady-state failure rates and does not specifically account for infant mortality failures (early-life failures) or wear out failures (end-of-life failures). It assumes a constant failure rate.

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6.0 SR-332, Method I: Unit Steady-State Failure Rate Parts Count Method: Factors

Method I: Unit Steady-State Failure Rate Using the Parts Count Method primarily estimates reliability by calculating the steady-state failure rate of a device using a set of generic parameters. It takes into account the following factors:

- 6.0.1 Number of component of each Type
- 6.0.2 Mean Generic Steady-State Failure Rate
- 6.0.3 Temperature factor
- 6.0.4 Environmental factor
- 6.0.5 Quality factor
- 6.0.6 Electrical Stress Factor



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7.0 ENVIRONMENTAL CONDITIONS

7.1 STORAGE AND OPERATIONAL ENVIRONMENT

The Reliability Prediction has been performed using **Ground, Fixed, Controlled (GB)** Environment based on the operational conditions of **Camera Model No.: HC-IPC-TI4312DVZK-N8-O**.

GROUND, FIXED, CONTROLLED (GB):

The environment represents a stationary, land-based installation located in a protected or climate-controlled facility. It is characterized by low vibration and shock stresses, low atmospheric variations, and low temperature-cycling stresses, with the understanding that a building or ruggedized enclosure provides consistent protection from weather and contaminants.

TEMPERATURE

Reliability Prediction has been performed on **Camera Model No.: HC-IPC-TI4312DVZK-N8-O** at the **Ambient Temperature of 40 °C** for **Ground, Fixed, Controlled (GB)** Environment.

7.2 QUALITY LEVEL : Level- 1

The unit has been evaluated under the commercial-grade component Quality level-1 as mentioned by the client in bill of material. The components used in the unit were procured without undergoing thorough device qualification or lot-to-lot controls by the equipment manufacturer.



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8.0 REPORTS

Report includes the following information-

- a) Components/ Parts descriptions
- b) Failure Rate
- c) Pi (π) Factors
- d) Parameters details

Refer Annexure- I.


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9.0 System Analysis Summary

The overall estimated MTBF and Failure Rate as per Telcordia SR-332 of **Camera Model No.: HC-IPC-TI4312DVZK-N8-O**, submitted for reliability prediction analysis by **Hi-Focus Electronics India Pvt Ltd** are summarized as follows:

SYSTEM: Camera Model No.: **HC-IPC-TI4312DVZK-N8-O**

Environment	Ambient Temperature	Failure Rate (Failures/Million Hours)	MTBF (Hrs)
Ground, Fixed, Controlled (GB)	40°C	4.117048	242892.49

- Design reliability prediction analysis was carried out based on the failure rates of components used in the camera (refer Annexure I), following a physical verification of components and in accordance with Telcordia SR-332, Issue 4.
- The camera contains three (03) nos of synchronous/stepper/shutter motors. However, failure rate data for these motors is not available in the Telcordia standard or in the reliability prediction software used. Therefore, for the purpose of the design reliability prediction, the failure rate for each motor has been conservatively assumed as one (1) FIT.
- MTBF (Mean Time Between Failures) calculations have been performed using the component details—such as type, quantity, electrical and environmental parameters, stress levels, and quality grades—as provided by the client in their submitted Bill of Materials.



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10.0 Abbreviation Used:

- a) MTBF : Mean Time Between Failure
- b) λ_G : Base failure rate for the generic part (Ground fixed base rate)
- c) σ_G : Standard deviation or uncertainty in base failure rate (generic)
- d) π_Q : Quality factor (related to part/process quality level)
- e) π_E : Environmental factor (based on the operating environment, e.g., ground benign, airborne)
- f) π_T : Temperature factor (influenced by ambient/operating temperature)
- g) π_S : Stress factor (e.g., voltage or power stress ratio)
- h) λ_{SS} : Subsystem or system-level failure rate (or stress-adjusted failure rate)
- i) σ_{SS} : Standard deviation or uncertainty in system/subsystem failure rate

- End of Report -


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Annexure - I



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CCTV CAMERA		Telcordia Issue 4	
Name	Make : HI-FOCUS	Calculation Model	Environment
Description	Model Number : HC-IPC-TI4312DVZK-N8-O	Ground, Fixed, Controlled (GB)	
	Serial Number : HFP25100543120730	Ambient Temperature, Celsius	
	Job No : 10TE202601266	40.00	
Quantity	1	Operating Profile	Telecommunications
Reference Designator		Failure Rate Units	Failures/Million Hours
		Failure Rate	4.117048
		MTBF Units	Hours
		MTBF 242892.49	

[illegible]

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.4	R0402RXX223 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R24	22K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.5	R0402RXX102 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R70,R95	1K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.6	R0402RXX105 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R27	1M $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: mohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.7	R0402RXX103 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R7,R11,R29,R30, R39,R93	10K $\pm$ 1%-0402-(-55°C~155°C)	6	0.000175	0.001048	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.8	R0402RXX104 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R31, R89, R96	100K $\pm$ 1%-0402-(-55°C~155°C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.9	R0402RXX241 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R38	2400 $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.10	R0402JXX000 OP16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R32, R34, R44	00 $\pm$ 5%-0402-(-55°C~155°C)	3	0.000152	4.562529E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.53301804, λ G: 0.08, σ G: 0.02, λ SS: 0.12792433, σ SS: 0.031981083, Temperature: 40
1.11	R0402RXX183 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R33, R37, R69, R88	18K $\pm$ 1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.12	R0402RXX152 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R92	1.5K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1.5, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.13	R0402RXX332 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R18	3.3K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.14	R0402RXX473 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R87	47K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 47, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.15	R0402RXX751 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R35	750 Ω $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 750, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.16	R0402RXX153 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R73	15K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.17	R0402RXX471 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R64, R100, R706	4700 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.18	R0402RXX474 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R5	470K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.19	R0402RXX472 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R42, R67, R151, R174, R274, R277, R431, R512, R513, R598, R703, R712, R737, R742, R746, R787, R788, R789, R790, R791, R792, R803, R809, R815, R818, R822, R823, R824, R859, R860, R871, R872, R873, R874, R886, R897, R1039, R1040	4.7K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	38	0.000162	0.006167	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.20	R0402RXX330 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R36,R75,R76,R7 7,R78,R87,R89,R 90,R91,R105,R1 81,R184,R187,R 202,R230,R231, R234,R514,R515 ,R804,R811,R85 0,R853,R854,R8 55,R856,R857,R 858,R864,R865, R866,R867,R868	33 Ω ±1%-0402-(-55°C~155°C)	33	0.000163	0.005374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.21	R0402RXX202 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R45,R63,R168,R 170,R173,R186, R243,R276,R299 ,R315,R728,R89 6	2K±1%-0402-(-55°C~155°C)	12	0.000169	0.002022	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.22	R0402RXX223 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R724	22K±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.23	R0402RXX222 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R61,R94,R563,R 730,R915	2.2K±1%-0402-(-55°C~155°C)	5	0.000177	8.836711E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.24	R0402RXX102 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R69,R345,R734, R744,R748,R793 ,R830,R1003,R1 005	1K $\pm$ 1%-0402-(- 55°C~155°C)	9	0.000171	0.001537	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Rated Power: 0.01, Applied Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.25	R0402RXX103 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R14,R39,R44,R7 3,R709,R722,R7 36,R738,R740,R 741,R794,R831, R1013,R1015,R1 016	10K $\pm$ 1%-0402-(- 55°C~155°C)	15	0.000167	0.002505	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.26	R0402RXX104 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R721,R801	100K $\pm$ 1%-0402-(- 55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.27	R0402RXX241 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1007,R1008	2400 $\pm$ 1%-0402-(- 55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 240, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.28	CR-05JL7---10R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R262, R899	100 $\pm$ 5%-0805-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40
1.29	CR-05FL7--100R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R242	100 $\pm$ 1%-0805-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000185	1.850080E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40
1.30	CR-05JL7--0R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R23	00 $\pm$ 5%-0805-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000168	1.684770E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.52750338, λ G: 0.08, σ G: 0.02, λ SS: 0.12660081, σ SS: 0.031650203, Temperature: 40
1.31	R0402JXX000 OP16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1, R2, R7, R9, R11, R15, R16, R17, R59, R66, R70, R82, R86, R88, R166, R546, R723, R727, R735, R739, R750, R819, R833, R912, R1001, R1002	00 $\pm$ 5%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	26	0.000136	0.003535	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.53301804, λ G: 0.08, σ G: 0.02, λ SS: 0.12792433, σ SS: 0.031981083, Temperature: 40
1.32	R0402RXX100 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R112	100 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.33	R0402RXX101 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R3,R4,R52,R101, R106,R814,R817	100Q $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.34	CR-03FL7--- 75R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R780,R783	75Q $\pm$ 1%-0603-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000566	0.001133	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 75, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 1.9155408, λ G: 0.08, σ G: 0.02, λ SS: 0.4597298, σ SS: 0.11493245, Temperature: 40
1.35	R0402RXX183 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R729	18K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.36	R0402RXX333 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R743,R745	33K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.37	R0402RXX332 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R717, R832, R898	3.3K $\pm$ 1%-0402-(-55°C~155°C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.38	R0402RXX751 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R726	750 Ω $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 750, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.39	R0402RXX153 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R704, R720, R725, R732, R747	15K $\pm$ 1%-0402-(-55°C~155°C)	5	0.000177	8.836711E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 15, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.40	R0402RWX2R 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8, R10, R12, R13	2.20 $\pm$ 1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.41	RS-05FL7--- OR2	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R80, R585	200mR $\pm$ 1%-0805-(-55°C~155°C)	2	0.000260	5.193472E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0.2, Resistance Units: ohm, Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.87809543, λ G: 0.08, σ G: 0.02, λ SS: 0.2107429, σ SS: 0.052685726, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.42	CR0402FF56R OG	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1014, R1017	56R±1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.43	CR0402FF680 OG	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1018, R1019, R1020, R1021	680R±1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.44	RL0805FR-070R3L	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R79, R584	300mR±1%-0805-(-55°C~155°C)	2	0.000260	5.193472E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.87809543, λ G: 0.08, σ G: 0.02, λ SS: 0.2107429, σ SS: 0.052685726, Temperature: 40
1.45	R0402RXX510 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R35	51Ω±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.46	R0402RXX474 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R37	470K±1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.47	R0402RXX223 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R26,R36	22K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.48	R0402RXX102 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R29	1K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.49	R0402RXX103 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R13,R22	10K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.50	CR-05FL7--- 10K	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R9,R10	10K $\pm$ 1%-0805-(-55°C~155°C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40



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Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.51	CR-051L7--100R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R2,R3	1000 $\pm$ 1%-0805-(-55°C~155°C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40
1.52	CR-051L7---0R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R17	00 $\pm$ 5%-0805-(-55°C~155°C)	1	0.000168	1.684770E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.52750338, λ G: 0.08, σ G: 0.02, λ SS: 0.12660081, σ SS: 0.031650203, Temperature: 40
1.53	RC0805JR-071RL	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R30,R32,R33,R34	10 $\pm$ 5%-0805-(-55°C~155°C)	4	0.000162	6.468270E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40
1.54	R0402RX000 OP16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8,R14	00 $\pm$ 5%-0402-(-55°C~155°C)	2	0.000158	3.152521E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.53301804, λ G: 0.08, σ G: 0.02, λ SS: 0.12792433, σ SS: 0.031981083, Temperature: 40
1.55	R0402RX100 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R21	10Q $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.56	R0402RXX183 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R20	18K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.57	R0402RXX333 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R15	33K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.58	R0402RXX332 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R6,R7,R11,R12, R27	3.3K $\pm$ 1%-0402-(-55°C~155°C)	5	0.000177	8.836711E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.59	R0402RXX473 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R19,R25	47K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 47, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.60	R0402RXX201 XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R28	200 Ω ±1%-0402- (-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 200, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.61	CC0402JRNPO 98N100	Capacitor	Ceramic	General	C81	10pF±5%-50V- NPO-0402	1	0.000055	5.526646E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16329449, λ G: 0.1, σ G: 0.01, λ SS: 0.048988348, σ SS: 0.0048988348, Temperature: 40
1.62	CC0402KRX7R 98B102	Capacitor	Ceramic	General	C47	1nF±10%-50V- X7R-0402	1	0.000048	4.815829E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, λ G: 0.1, σ G: 0.01, λ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40
1.63	0402X105X6R 3CT	Capacitor	Ceramic	General	C30,C31,C32,C58	1uF±10%-6.3V- X5R-0402	4	0.000090	3.589336E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.28109921, λ G: 0.1, σ G: 0.01, λ SS: 0.084329762, σ SS: 0.0084329762, Temperature: 40
1.64	CC0402KRX7R 78B104	Capacitor	Ceramic	General	C14,C33,C34,C38,C54,C56,C59,C60,C61,C62,C63,C64,C70,C80,C88,C89,C93,C99,C457	100nF±10%- 16V-X7R-0402	19	0.000093	0.001760	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.65	0402B103X50 0CT	Capacitor	Ceramic	General	C11,C13,C20,C44	10nF±10%-50V- X7R-0402	4	0.000045	1.813783E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, λ G: 0.1, σ G: 0.01, λ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.66	0402X225M6 R3CT	Capacitor	Ceramic	General	C12	2.2uF±20%- 6.3V-XSR-0402	1	0.000288	2.876279E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, λ G: 0.1, σ G: 0.01, λ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.67	CL05A475MQ 5NRNC	Capacitor	Ceramic	General	C3,C6,C7,C9,C1 5,C17,C19,C21,C 23,C25,C27,C36	4.7 uF±20%- 6.3V-XSR-0402	12	0.000264	0.003173	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, λ G: 0.1, σ G: 0.01, λ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.68	0402X224K10 ONT	Capacitor	Ceramic	General	C86,C104	220nF±20%- 10V-XSR-0402	2	0.000163	3.259277E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 10, Percentage: 33	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 10, Percentage: 33	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.4980773, λ G: 0.1, σ G: 0.01, λ SS: 0.14942319, σ SS: 0.014942319, Temperature: 40
1.69	CL10A106KP8 NNNC	Capacitor	Ceramic	General	C39,C57,C71,C4 36	10uF±10%-10V- XSR-0603	4	0.000319	0.001277	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	π Q: 3, π E: 1, π T: 1, π S: 1, λ G: 0.1, σ G: 0.01, λ SS: 0.3, σ SS: 0.03, Temperature: 40
1.70	CL10A226MQ 8NRNC	Capacitor	Ceramic	General	C37,C40,C42,C5 1,C55,C79,C90,C 92,C94,C96,C10 3,C105,C107,C2 32,C233	22uF±20%-6.3V- XSR-0603	15	0.000342	0.005126	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	π Q: 3, π E: 1, π T: 1, π S: 1.1025427, λ G: 0.1, σ G: 0.01, λ SS: 0.33076281, σ SS: 0.033076281, Temperature: 40
1.71	CC0402JRNPO 9BN100	Capacitor	Ceramic	General	C723,C825	10pF±5%-50V- NPO-0402	2	0.000053	1.068553E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16329449, λ G: 0.1, σ G: 0.01, λ SS: 0.048988348, σ SS: 0.0048988348, Temperature: 40



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1.72	CC0402JRNPO 98N180	Capacitor	Ceramic	General	C815,C822,C824	18pF±5%-50V- NPO-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.16873936, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.050621808,$ $\sigma SS: 0.0050621808,$ Temperature: 40
1.73	0402N270J50 OCT	Capacitor	Ceramic	General	C23	27pF-50V-NPO- ±5%-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.16873936, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.050621808,$ $\sigma SS: 0.0050621808,$ Temperature: 40
1.74	0402N330J50 OCT	Capacitor	Ceramic	General	C12,C18,C31,C6 4	33pF-50V-NPO- ±5%-0402	4	0.000054	2.154621E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.16873936, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.050621808,$ $\sigma SS: 0.0050621808,$ Temperature: 40
1.75	CC0402JRNPO 98N101	Capacitor	Ceramic	General	C17	100pF±5%-50V- NPO-0402	1	0.000065	6.576536E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 50, Percentage: 10	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 50, Percentage: 10	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.19398004, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.058194013,$ $\sigma SS: 0.0058194013,$ Temperature: 40
1.76	CC0402KRX7R 98B102	Capacitor	Ceramic	General	C8,C11,C32,C33, C52,C76	1nF±10%-50V- X7R-0402	6	0.000045	2.690611E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.14204662, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.042613985,$ $\sigma SS: 0.0042613985,$ Temperature: 40
1.77	0402B332K50 OCT	Capacitor	Ceramic	General	C93,C489	3.3nF±10%-50V- X7R-0402	2	0.000055	1.104183E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.16873936, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.050621808,$ $\sigma SS: 0.0050621808,$ Temperature: 40
1.78	CC1206KKX7R DBB102	Capacitor	Ceramic	General	C6,C260,C301,C 550	1nF±10%-2KV- X7R-1206	4	0.000042	1.660743E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.13006122, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda SS: 0.039018367,$ $\sigma SS: 0.0039018367,$ Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.79	0402N221J50 OCT	Capacitor	Ceramic	General	C705, C716, C731	220pF $\pm$ 5%-50V- NPO-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.16873936, λG : 0.1, σG : 0.01, λSS : 0.050621808, σSS : 0.0050621808, Temperature: 40
1.80	CC0805KKX5R 788106	Capacitor	Ceramic	General	C15, C77, C277, C 331	10uF $\pm$ 10%-16V- X5R-0805	4	0.000096	3.829139E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.29987937, λG : 0.1, σG : 0.01, λSS : 0.08996381, σSS : 0.008996381, Temperature: 40
1.81	0603N821J50 OCT	Capacitor	Ceramic	General	C381, C382	820pF $\pm$ 5%-50V- NPO-0603	2	0.000055	1.104183E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.16873936, λG : 0.1, σG : 0.01, λSS : 0.050621808, σSS : 0.0050621808, Temperature: 40
1.82	0402X105K6R 3CT	Capacitor	Ceramic	General	C739, C801, C821	1uF $\pm$ 10%-6.3V- X5R-0402	3	0.000091	2.717081E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.28109921, λG : 0.1, σG : 0.01, λSS : 0.084329762, σSS : 0.0084329762, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	n Factors
1.83	CC0402KRX7R 788104	Capacitor	Ceramic	General	C5,C9,C10,C19, C28,C40,C44,C5 7,C66,C70,C86,C 119,C137,C253, C254,C276,C282 ,C334,C711,C71 5,C719,C726,C7 33,C738,C741,C 747,C751,C753, C802,C804,C806 ,C810,C811,C81 3,C814,C817,C8 18,C819,C820,C 823,C827,C829, C830,C831,C832 ,C835,C836,C85 0,C852,C853,C8 54,C855,C858,C 859,C860,C998, C1002,C1003,C1 004,C1006,C101 2,C1013,C1014, C1015,C1016,C1 017,C1018,C101 9,C1020,C1021, C1022,C1023,C1 024,C1025,C102 6,C1028,C1029, C1030,C1031,C1 032,C1033,C103 4,C1035,C1036, C1037,C1038,C1 039,C1040,C104 1,C1102	100nF±10%- 16V-X7R-0402	90	0.000091	0.008206	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.29987937, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda S: 0.08996381$, $\sigma S: 0.008996381$, Temperature: 40
1.84	0402B103K50 OCT	Capacitor	Ceramic	General	C56,C1042	10nF±10%-50V- X7R-0402	2	0.000046	9.295127E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S:$ 0.14204662, $\lambda G: 0.1, \sigma G:$ 0.01, $\lambda S: 0.042613985$, $\sigma S: 0.0042613985$, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.85	0402X225M5 R3CT	Capacitor	Ceramic	General	C24,C98,C816	2.2uF±20%- 6.3V-X5R-0402	3	0.000274	8.214552E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, Δ G: 0.1, σ G: 0.01, Δ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.86	0805X226M1 00CT	Capacitor	Ceramic	General	C69	22uF±20%-10V- 0805	1	0.000338	3.384465E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	π Q: 3, π E: 1, π T: 1, π S: Δ G: 0.1, σ G: 0.01, Δ SS: 0.3, σ SS: 0.03, Temperature: 40
1.87	UT1E101M06 07VG	Capacitor	Electrolytic	Aluminum	CE1	100uF±20%- 25V-105°C- Ø6.3*7.7mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 100, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.95313379, Δ G: 0.73, σ G: 0.17, Δ SS: 2.087363, σ SS: 0.48609823, Temperature: 40
1.88	CL05A475MQ 5NRNC	Capacitor	Ceramic	General	C73,C75,C186,C 805,C807,C808, C809,C828,C857	4.7uF±20%- 6.3V-X5R-0402	9	0.000266	0.002393	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, Δ G: 0.1, σ G: 0.01, Δ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.89	CL10A106KP8 NNNC	Capacitor	Ceramic	General	C1,C16,C41,C29 6,C704,C714,C7 21,C728,C735,C 746,C812,C826, C851,C856,C100 1,C1027	10uF±10%-10V- X5R-0603	16	0.000310	0.004954	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	π Q: 3, π E: 1, π T: 1, π S: Δ G: 0.1, σ G: 0.01, Δ SS: 0.3, σ SS: 0.03, Temperature: 40
1.90	CL21A106KAY NNNE	Capacitor	Ceramic	General	C289,C333,C743	10uF±10%-25V- X5R-0805	3	0.000297	8.904935E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.92127196, Δ G: 0.1, σ G: 0.01, Δ SS: 0.27638159, σ SS: 0.027638159, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.91	CL10A226MQ 8NRNC	Capacitor	Ceramic	General	C20,C21,C197,C 706,C710,C717, C730,C732,C737 ,C750,C803	22 μ F $\pm$ 20%-6.3V- X5R-0603	11	0.000344	0.003779	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	π Q: 3, π E: 1, π T: 1, π S: 1.1025427, λ G: 0.1, σ G: 0.01, λ SS: 0.33076281, σ SS: 0.033076281, Temperature: 40
1.92	0402N270J50 OCT	Capacitor	Ceramic	General	C8	27pF-50V-NPO- $\pm$ 5%-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.93	CC1206KKX7R DB8102	Capacitor	Ceramic	General	C7,C9,C10	1nF $\pm$ 10%-2KV- X7R-1206	3	0.000042	1.257161E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.13006122, λ G: 0.1, σ G: 0.01, λ SS: 0.039018367, σ SS: 0.0039018367, Temperature: 40
1.94	0402N221J50 OCT	Capacitor	Ceramic	General	C17,C19	220pF $\pm$ 5%-50V- NPO-0402	2	0.000055	1.104183E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.95	1206B104K20 1NT	Capacitor	Ceramic	General	C2,C3,C12	100nF $\pm$ 10%- 200V/250V- X7R-1206	3	0.000111	3.328779E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.3443833, λ G: 0.1, σ G: 0.01, λ SS: 0.10331499, σ SS: 0.010331499, Temperature: 40
1.96	1206B105K10 1CT	Capacitor	Ceramic	General	C5	1uF $\pm$ 10%-100V- X7R-1206	1	0.000312	3.123403E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.92127196, λ G: 0.1, σ G: 0.01, λ SS: 0.27638159, σ SS: 0.027638159, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.97	VZH221M1ET R-0810	Capacitor	Electrolytic	Aluminum	CE2	220uF±20% 25V-105°C- Ø8*10mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 220, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.95313379, λG : 0.73, σG : 0.17, λS : 2.087363, σS : 0.48609823, Temperature: 40
1.98	CC0402KRX7R 7B8104	Capacitor	Ceramic	General	C11	100nF±10% 16V-X7R-0402	1	0.000102	1.016686E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 16, Rated Voltage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.29987937, λG : 0.1, σG : 0.01, λS : 0.08996381, σS : 0.008996381, Temperature: 40
1.99	UL1J470W081 0VG	Capacitor	Electrolytic	Aluminum	CE1	47uF±20%-63V- 105°C- Ø8*10mm	1	0.005369	0.005369	Type: Aluminum, Quality Level: Level I, Capacitance: 47, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	Level I	πQ : 3, πE : 1, πT : 1, πS : 1.8749302, λG : 0.73, σG : 0.17, λS : 4.1060971, σS : 0.9562144, Temperature: 40
1.100	CL21A106KAY NNNE	Capacitor	Ceramic	General	C6,C14,C15	10uF±10%-25V- X5R-0805	3	0.000297	8.904935E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.92127196, λG : 0.1, σG : 0.01, λS : 0.27638159, σS : 0.027638159, Temperature: 40
1.101	TCC0805X7R1 02K201DT	Capacitor	Ceramic	General	C21	1nF±20%-200V- X7R-0805	1	0.000056	5.572151E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 200, Percentage: 6	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 200, Percentage: 6	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.16463901, λG : 0.1, σG : 0.01, λS : 0.049391704, σS : 0.0049391704, Temperature: 40
1.102	11-1832002- 330	Connection	Multi-Pin		J4	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	λG : 0.08, σG : 0.02, πQ : 3, πE : 1, πT : 1, λS : 0.24, σS : 0.06, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.103	1.25T-1-WT-4A	Connection	Multi-Pin		J2	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	λG : 0.16, σG : 0.04, πQ : 3, πE : 1, πT : 1, λSS : 0.48, σSS : 0.12, Temperature: 40
1.104	60076-5007-1001	Connection	Multi-Pin		J5	50*0.5mm-Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λG : 2, σG : 0.5, πQ : 3, πE : 1, πT : 1, λSS : 6, σSS : 1.5, Temperature: 40
1.105	C1402-1009-211	Connection	Multi-Pin		J2	193-F093K230-B06	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	λG : 0.4, σG : 0.1, πQ : 3, πE : 1, πT : 1, λSS : 1.2, σSS : 0.3, Temperature: 40
1.106	1.25T-1-LT-10A	Connection	Multi-Pin		J9	10*1.25mm	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	λG : 0.4, σG : 0.1, πQ : 3, πE : 1, πT : 1, λSS : 1.2, σSS : 0.3, Temperature: 40
1.107	1.25T-1-LT-8A	Connection	Multi-Pin		J6	8*1.25mm	1	0.001278	0.001278	Quality Level: Level I, Number of Pins: 8		Level I	λG : 0.32, σG : 0.08, πQ : 3, πE : 1, πT : 1, λSS : 0.96, σSS : 0.24, Temperature: 40
1.108	1.25T-1-LT-6A	Connection	Multi-Pin		J5	6*1.25mm	1	0.000958	9.581568E-004	Quality Level: Level I, Number of Pins: 6		Level I	λG : 0.24, σG : 0.06, πQ : 3, πE : 1, πT : 1, λSS : 0.72, σSS : 0.18, Temperature: 40
1.109	1.25T-1-LT-4A	Connection	Multi-Pin		J3	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	λG : 0.16, σG : 0.04, πQ : 3, πE : 1, πT : 1, λSS : 0.48, σSS : 0.12, Temperature: 40
1.110	1.25T-1-LT-2A	Connection	Multi-Pin		J8	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	λG : 0.08, σG : 0.02, πQ : 3, πE : 1, πT : 1, λSS : 0.24, σSS : 0.06, Temperature: 40
1.111	1.25T-1-LT-12A	Connection	Multi-Pin		J1	12*1.25mm	1	0.001916	0.001916	Quality Level: Level I, Number of Pins: 12		Level I	λG : 0.48, σG : 0.12, πQ : 3, πE : 1, πT : 1, λSS : 1.44, σSS : 0.36, Temperature: 40
1.112	60076-5007-1001	Connection	Multi-Pin		J11	50*0.5mm-Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λG : 2, σG : 0.5, πQ : 3, πE : 1, πT : 1, λSS : 6, σSS : 1.5, Temperature: 40
1.113	1.25T-1-WT-6A	Connection	Multi-Pin		J5	6*1.25mm	1	0.000958	9.581568E-004	Quality Level: Level I, Number of Pins: 6		Level I	λG : 0.24, σG : 0.06, πQ : 3, πE : 1, πT : 1, λSS : 0.72, σSS : 0.18, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.114	1.25T-1-LT-4A	Connection	Multi-Pin		J1	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	λG : 0.16, σG : 0.04, πQ : 3, πE : 1, πT : 1, λS : 0.48, σS : 0.12, Temperature: 40
1.115	NRS3015T2R2 MINGH	Inductor	Coil	Chip/Ceramic Filter	L702,L703,L704	2.2uH-1.48A	3	0.000462	0.001386	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λS : 0.3, σS : 0.21, Temperature: 40
1.116	HXNR4030B-4R7M	Inductor	Coil	Chip/Ceramic Filter	L1,L2,L705	4.7uH-2A	3	0.000462	0.001386	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λS : 0.3, σS : 0.21, Temperature: 40
1.117	0.47mH-1.8A	Inductor	Coil	Chip/Ceramic Filter	L2	0.47mH-1.8A	1	0.000581	5.806969E-004	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λS : 0.3, σS : 0.21, Temperature: 40
1.118	P0080SB-LV1	Semiconductor	Diode	General Purpose, Si	D1,D2,D9,D31,D85,D86	6.5V-4KV-SMB	6	0.004517	0.027101	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.001, Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.23566226, λG : 4.6, σG : 3.3, λS : 3.2521391, σS : 2.3330563, Temperature: 40
1.119	1SS355	Semiconductor	Diode	General Purpose, Si	D11,D69,D801	80V-0.1A-SOD323	3	0.000477	0.001432	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.01, Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.31348618, λG : 0.33, σG : 0.23, λS : 0.31035132, σS : 0.21630546, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.120	SCD310H	Semiconductor	Diode	General Purpose, Si	D3,D4	100V-3A-2010/SMAF	2	0.001014	0.002028	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.61672421, λG : 0.33, σG : 0.23, λSS : 0.61055697, σSS : 0.42553971, Temperature: 40
1.121	BV- SMBJ20CAI	Semiconductor	Diode	General Purpose, Si	D5	20V-10/1000uS-800W-30V@8/20uS-SMB	1	0.006982	0.006982	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.25837757, λG : 4.6, σG : 3.3, λSS : 3.5656104, σSS : 2.5579379, Temperature: 40
1.122	TUSD03FB	Semiconductor	Diode	TVS Surge Suppressor	D28,D29	3.3V-8/20uS-350W-Max19V-SOD323	2	0.002327	0.004655	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.001, Rated Forward Current: 8, Percentage: 0.0125	Rated Forward Current: 8, Percentage: 0.0125	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.23465534, λG : 2.55, σG : 0.81, λSS : 1.7951133, σSS : 0.57021246, Temperature: 40
1.123	RS1M-F1-0002	Semiconductor	Diode	General Purpose, Si	D3	1000V-1A-SMA	1	0.000801	8.012066E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.2, Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.41895155, λG : 0.33, σG : 0.23, λSS : 0.41476203, σSS : 0.28907657, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.124	1SS355	Semiconductor	Diode	General Purpose, Si	D4	80V-0.1A-SOD323	1	0.000600	5.995137E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.01, Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.31348618, λ G: 0.33, σ G: 0.23, λ SS: 0.31035132, σ SS: 0.21630546, Temperature: 40
1.125	SCD310H	Semiconductor	Diode	General Purpose, Si	D7	100V-3A-2010/SMAF	1	0.001179	0.001179	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.61672421, λ G: 0.33, σ G: 0.23, λ SS: 0.61055697, σ SS: 0.42553971, Temperature: 40
1.126	BV-SMBJ58C2H	Semiconductor	Diode	TVS Surge Suppressor	D2,D5	58V-10us/700us-85MAX-SMB	2	0.002390	0.004781	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.2, Rated Forward Current: 21.37, Percentage: 0.93589143659	Rated Forward Current: 21.37, Percentage: 0.93589143659	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.24102391, λ G: 2.55, σ G: 0.81, λ SS: 1.8438329, σ SS: 0.5856881, Temperature: 40
1.127	SS510B-F1-0000HF	Semiconductor	Diode	General Purpose, Si	D1	100V-5A-SMB	1	0.000801	8.012066E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 5, Percentage: 20, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 5, Percentage: 20, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.41895155, λ G: 0.33, σ G: 0.23, λ SS: 0.41476203, σ SS: 0.28907657, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.128	MB6F	Semiconductor	Diode	General Purpose, Ge	Z1,Z2	Glass Passivated Single-Phase Bridge Rectifier	2	0.074260	0.148520	Type: General Purpose, Ge, Quality Level: Level I, Operating Current: 0.4, Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 2.3869109, \lambda G: 6.2, \sigma G: 4.4, \lambda SS: 44.396542, \sigma SS: 31.507223$, Temperature: 40
1.129	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.33333333333	Rated Power: 0.3, Percentage: 3.33333333333	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.32627979, \lambda G: 3.44, \sigma G: 0.75, \lambda SS: 3.3672075, \sigma SS: 0.73412954$, Temperature: 40
1.130	S8550	Semiconductor	Transistor	Field Effect, Silicon	Q3	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.33333333333	Rated Power: 0.3, Percentage: 3.33333333333	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.32627979, \lambda G: 3.44, \sigma G: 0.75, \lambda SS: 3.3672075, \sigma SS: 0.73412954$, Temperature: 40
1.131	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q4,Q7,Q14	40V-0.5A-SOT23	3	0.003921	0.011762	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.33333333333	Rated Power: 0.3, Percentage: 3.33333333333	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.32627979, \lambda G: 3.44, \sigma G: 0.75, \lambda SS: 3.3672075, \sigma SS: 0.73412954$, Temperature: 40
1.132	VJ52305A-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q8	20V-5.4A-SOT23-6L	1	0.007293	0.007293	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.3, Rated Power: 1.2, Percentage: 25	Rated Power: 1.2, Percentage: 25	Level I	$\pi Q: 3, \pi E: 1, \pi T: 1, \pi S: 0.54881164, \lambda G: 3.44, \sigma G: 0.75, \lambda SS: 5.6637361, \sigma SS: 1.2348262$, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.133	BSS138W-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q3	50V-0.34A-SOT323	1	0.004697	0.004697	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.15, Percentage: 6.6666666667	Rated Power: 0.15, Percentage: 6.6666666667	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.35345468, λG : 3.44, σG : 0.75, λSS : 3.6476523, σSS : 0.79527303, Temperature: 40
1.134	S8550-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.3333333333	Rated Power: 0.3, Percentage: 3.3333333333	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.32627979, λG : 3.44, σG : 0.75, λSS : 3.3672075, σSS : 0.73412954, Temperature: 40
1.135	NCE60P04SN	Semiconductor	Transistor	Field Effect, Silicon	Q3	60V-4A-SOT23-6L	1	0.008907	0.008907	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.5, Rated Power: 1.5, Percentage: 33.3333333333	Rated Power: 1.5, Percentage: 33.3333333333	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.67032005, λG : 3.44, σG : 0.75, λSS : 6.9177029, σSS : 1.5082201, Temperature: 40
1.136	Light board	Optical Device	Other	Single LED / LCD Segment	LD1,LD2,LD3,L4	Night Mode	4	0.001110	0.004440	Type: Single LED / LCD Segment, Quality Level: Level I		Level I	πQ : 3, πE : 1, πT : 1, λG : 0.25, σG : 0.18, λSS : 0.75, σSS : 0.54, Temperature: 40
1.137	[HFD42/4.5-3SR(825)]-DC5V-HF	Relay	General Purpose		K1	HFD42/4.5-3SR(825)	1	0.009483	0.009483	Quality Level: Level I, Contact Current: 1, Rated Current: 4, Percentage: 25	Contact Current: 1, Rated Current: 4, Percentage: 25	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.72252735, λG : 1.9, σG : 1.9, λSS : 4.1184059, σSS : 4.1184059, Temperature: 40
1.138	T5TP650K-180BH-8.4	Switch	Pushbutton		SW1	DC12V-50mA-4PIN-6*6*5mm	1	0.007842	0.007842	Quality Level: Level I, Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.67705687, λG : 1.99, σG : 1.4, λSS : 4.0420295, σSS : 2.8436389, Temperature: 40



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Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.139	K-F-24.000M-9-2020-4085-80	Miscellaneous	Crystal Oscillator		X803	24M $\pm$ 20ppm-(-40°C-85°C)-3.2*2.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	πQ : 3, πE : 1, πT : 1, λG : 3.2, σG : 1.5, λSS : 9.6, σSS : 4.5, Temperature: 40
1.140	SF32WK32768 D31T002	Miscellaneous	Crystal Oscillator		X802	32.768K $\pm$ 20ppm-(-40°C-85°C)-3.2*1.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	πQ : 3, πE : 1, πT : 1, λG : 3.2, σG : 1.5, λSS : 9.6, σSS : 4.5, Temperature: 40
1.141	YT30028FB.IR-IZ-DK-	Rotating Device				Lens-Electric Zoom	2	1.000000	2.000000				
1.142	IR0109	Rotating Device				Day and Night Mode	1	1.000000	1.000000				
1.143	ETA5052S2G	Integrated Circuit	Analog		U2	[ETA5052S2G/E TA5052V0S2G]-SOT23-6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40
1.144	LP3981AB5F]-SOT23-5	Integrated Circuit	Analog		U3,U7	[LP3981AB5F]-SOT23-5	2	0.026871	0.053741	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40
1.145	[TP10-2-VR]-VSSOP8	Integrated Circuit	Analog		U4	[TP10-2-VR]-VSSOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40
1.146	[TPM16050-S6TR/TM18113]-SOT23-6	Integrated Circuit	Analog		U5	[TPM16050-S6TR/TM18113]-SOT23-6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40
1.147	TP8485E-SR	Integrated Circuit	Analog		U5	[TP8485E-SR/SIT3088EESA]-SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40



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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.148	DIO54335TST 6;-SOT23-6- dioo	Integrated Circuit	Analog		U709	[DIO54335TST6] -SOT23-6	1	0.028473	0.028473	Quality Level: Level I, Number of Transistors: 456		Level I	πQ : 3, πE : 1, πT : 1, λG : 4.8804343, σG : 3.450467, λSS : 14.641303, σSS : 10.351401, Temperature: 40
1.149	ETA3485S2F]- SOT23-5-ETA	Integrated Circuit	Analog		U702,U703,U70 6	[ETA3485S2F]- SOT23-5	3	0.022631	0.067893	Quality Level: Level I, Number of Transistors: 456		Level I	πQ : 3, πE : 1, πT : 1, λG : 4.8804343, σG : 3.450467, λSS : 14.641303, σSS : 10.351401, Temperature: 40
1.150	ETA5053V180 DF1E]-1.8V- DFN1*1-ETA	Integrated Circuit	Analog		U710	[ETA5053V180D F1E]-1.8V- DFN1*1	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40
1.151	LP3981AB5F]- SOT23-5	Integrated Circuit	Analog		U707	[LP3981AB5F]- SOT23-5	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40
1.152	TMI5302]- TSOT23-6	Integrated Circuit	Analog		U18,U38	[TMI5302]- TSOT23-6	2	0.030756	0.061513	Quality Level: Level I, Number of Transistors: 768		Level I	πQ : 3, πE : 1, πT : 1, λG : 6.1386501, σG : 4.3400256, λSS : 18.41595, σSS : 13.020077, Temperature: 40
1.153	MS41908M	Integrated Circuit	Analog					#DIV/0!	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40
1.154	[SD4952CTR]- SO16	Integrated Circuit	Analog		U1	[SD4952CTR]- SO16	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	πQ : 3, πE : 1, πT : 1, λG : 5.3630921, σG : 3.7917061, λSS : 16.089276, σSS : 11.375118, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n * Failure Rate	Part Data	Stress Data	Quality	π Factors
1.155	IMX307LQR-C	Integrated Circuit	Digital		U1	[IMX307LQR-C]- 2.0MP-1/2.8- LGA110	1	0.031007	0.031007	Quality Level: Level I, Technology: CMOS, Number of Gates: 5000000		Level I	π Q: 3, π E: 1, π T: 1, λ G: 6.9374105, σ G: 2.5470825, λ SS: 20.812232, σ SS: 7.6412476, Temperature: 40
1.156	W25N01GVZE IG	Integrated Circuit	ROM, PROM, EPROM		U1101	[W25N01GVZE] G]- WSON8 (8x6mm)-3V3	1	0.249449	0.249449	Quality Level: Level I, Technology: CMOS, Number of Bits: 1, Bit Units: Gb		Level I	π Q: 3, π E: 1, π T: 1, λ G: 42.756504, σ G: 30.228848, λ SS: 128.26951, σ SS: 90.686545, Temperature: 40
1.157	NT5CB256M1 6ER-FL	Integrated Circuit	RAM		U1001	[NT5CB256M16 ER-FL]-FBGA96	1	0.115747	0.115747	Quality Level: Level I, Technology: Dynamic NMOS/CMOS, Number of Bits: 4, Bit Units: Gb		Level I	π Q: 3, π E: 1, π T: 1, λ G: 20.379802, σ G: 13.601017, λ SS: 61.139406, σ SS: 40.80305, Temperature: 40
1.158	C5-36GW	Integrated Circuit	Microprocessor		U801	[C5-36GW]- BGA307	1	0.154317	0.154317	Quality Level: Level I, Technology: CMOS, Bus Width in Bits: 32		Level I	π Q: 3, π E: 1, π T: 1, λ G: 24.604507, σ G: 20.224777, λ SS: 73.81352, σ SS: 60.674332, Temperature: 40
1.159	S16403	Inductor	Transformer	Power (> 1W)	T1	[S16403]100M POE-1 channel	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λ G: 5, σ G: 2, π Q: 3, π E: 1, π T: 1, λ G: 5, σ G: 2, λ SS: 15, σ SS: 6, Temperature: 40
1.160	GEP13	Inductor	Transformer	Power (> 1W)	T1	GEP13-127 μ H	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λ G: 5, σ G: 2, π Q: 3, π E: 1, π T: 1, λ G: 5, σ G: 2, λ SS: 15, σ SS: 6, Temperature: 40
1.161	[ML414H- IV01E]-3V	Miscellaneous	Battery		BAT801	Rechargeable, 1mAh	1	0.104891	0.104891	Battery Type: Lithium, Quality Level: Level I		Level I	π Q: 3, π E: 1, λ G: 25, σ G: 7.5, λ SS: 75, σ SS: 22.5
1.162	Fuse-4A	Miscellaneous	Fuse		FU1	[6125S8A250V -G]-4A-250V 6125	1	0.001964	0.001964	Fuse Type: $\leq$ 30A, Quality Level: Level I		Level I	π Q: 3, π E: 1, λ G: 0.4, σ G: 0.19, λ SS: 1.2, σ SS: 0.57



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